Significance of the mean closeness centrality in syntactic dependency networks

[TODO: names]
Complex and Social Networks (2026–2027), MIRI

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1 Introduction

We study the global syntactic dependency network of ten languages from the Universal Dependencies (UD) collection. Vertices are words and two words are linked if they formed a syntactic dependency at least once in the treebanks of the language. We use the undirected version of these networks.

The metric of interest is the mean closeness centrality

$$\mathcal{C} = \frac{1}{N}\sum_{i=1}^{N} \mathcal{C}_i, \qquad \mathcal{C}_i = \frac{1}{N-1}\sum_{\substack{j=1 \\ j\neq i}}^{N} \frac{1}{d_{ij}}, \tag{1}$$

where d_{ij} is the geodesic distance between i and j (the term is 0 when j is unreachable from i, so all $\mathcal{C}_i$ are finite and $0 \leq \mathcal{C}_i \leq 1$).

We ask whether $\mathcal{C}$ is *significantly large*, i.e. whether $p(\mathcal{C}_{NH} \geq \mathcal{C})$ is small, under two null hypotheses: a binomial (Erdős–Rényi) graph with the same N and E, and a random graph with the same degree sequence as the real network (switching model). The p-values are estimated with a Monte Carlo procedure. [TODO: one or two sentences announcing the main finding once results are in.]

2 Results

2.1 Basic properties of the networks

Table **??** summarises the networks after removing loops and repeated edges. N is the number of vertices, E the number of edges, $\langle k \rangle = 2E/N$ the mean degree and $\delta = 2E/(N(N-1))$ the network density.

2.2 Significance of the mean closeness centrality

Table **??** shows the value of $\mathcal{C}$ in each real network and the estimated p-values under the two null hypotheses.

3 Discussion

[TODO: Summary of the results and interpretation. Points the assignment asks for:]

- Under which null hypothesis is $\mathcal{C}$ significantly large and under which is it not? Reason or speculate why (e.g. the role of hubs and of the degree sequence).

Language	N	E	$\langle k \rangle$	δ
Arabic	27563	176829	12.83	$4.66 \cdot 10^{-4}$
Basque	24063	74902	6.23	$2.59 \cdot 10^{-4}$
Catalan	32298	254087	15.73	$4.87 \cdot 10^{-4}$
Chinese	24827	106915	8.61	$3.47 \cdot 10^{-4}$
Czech	215082	1925887	17.91	$8.33 \cdot 10^{-5}$
English	38022	406610	21.39	$5.63 \cdot 10^{-4}$
Greek	18695	68967	7.38	$3.95 \cdot 10^{-4}$
Hungarian	13456	31406	4.67	$3.47 \cdot 10^{-4}$
Italian	52305	399894	15.29	$2.92 \cdot 10^{-4}$
Turkish	100790	384538	7.63	$7.57 \cdot 10^{-5}$

Table 1: Summary of the properties of the networks.

Language	$\mathcal{C}$	p-value (binomial)	p-value (switching)	mode
Arabic	0.3174	0.00	1.00	exact, $T = 100$
Basque	0.2686	0.00	0.59	exact, $T = 100$
Catalan	0.3409	0.00	1.00	exact, $T = 100$
Chinese	0.2915	0.00	0.17	exact, $T = 100$
Czech	0.3057	0.00	1.00	approx. ($M/N = 1\%$, $T = 50$)
English	0.3364	0.00	1.00	approx. ($M/N = 10\%$, $T = 100$)
Greek	0.2867	0.00	1.00	exact, $T = 100$
Hungarian	0.2637	0.00	1.00	exact, $T = 100$
Italian	0.3271	0.00	1.00	approx. ($M/N = 10\%$, $T = 100$)
Turkish	0.2643	0.00	1.00	approx. ($M/N = 10\%$, $T = 100$)

Table 2: Mean closeness centrality of each network (computed exactly) and estimated p-values. T is the number of random graphs per null model; "approx." means that the random graphs' $\mathcal{C}_{NH}$ was estimated from a fraction M/N of their vertices. A p-value of 0.00 means that none of the T random graphs reached the real value.

- To what extent do languages resemble or differ in the value of $\mathcal{C}$ and in the outcome of the significance test? Relate to N, $\langle k \rangle$ and δ (Table **??**).
- Conclusions, and the limitations coming from approximate computation for the large languages.

4 Methods

4.1 Network construction

The networks are built from the UD treebanks v2.18 for the same ten languages as in Lab 2 (script `code/build_networks.R`). For each language, all its treebanks are pooled, except four that were excluded: UD_Arabic-NYUAD and UD_English-GUMReddit (no word forms), UD_Italian-Old (historical language) and UD_Chinese-GSDSimp (same text as Chinese-GSD).

- Vertices are lowercased word forms. In the files, the two words of an edge are separated by a tab, since a few forms contain spaces; the program checks the header (N, E) against the file when loading it.
- For every token whose head is not the root, the pair {head, dependent} is an edge. Edges involving punctuation (UPOS `PUNCT`) are ignored.

- Direction is discarded: each pair is stored once in canonical order, so that head→dependent and dependent→head are the same edge.
- Loops are removed and repeated edges are counted once. Pairs are put in canonical order with byte-wise (locale-independent) string comparison.

N counts the words that appear in at least one edge. Each network is written to `data/language_dependency_` (first line: N E; then one edge per line). Note that these networks are unions of many sentence trees; they are not trees and contain cycles.

4.2 Computing the mean closeness centrality

The network is stored as an adjacency list in compressed sparse row form. $\mathcal{C}_i$ is obtained with a breadth-first search (BFS) from i, adding $1/d_{ij}$ for each vertex reached. Isolated vertices have $\mathcal{C}_i = 0$ and need no search. The code is written in C++17 and compiled with `-O3`; BFS runs, and independent random replicates, are parallelised with OpenMP (8 threads).

4.3 Null models and Monte Carlo estimation

The p-value is estimated as $p(\mathcal{C}_{NH} \geq \mathcal{C}) \approx f/T$, where T is the number of random graphs generated and f the number of them with $\mathcal{C}_{NH} \geq \mathcal{C}$. Comparisons use a tolerance of 10^{-12} to avoid floating-point order effects. We use $T = 100$ random graphs per null model, except for Czech ($T = 50$); the values are listed in Table **??**.

Binomial graph. A graph with N vertices and E edges: E distinct pairs of distinct vertices are drawn uniformly, rejecting loops and repeated pairs.

Switching model. Starting from the real edge list, we perform QE attempts. Each attempt picks two edges $u \sim v$ and $s \sim t$ uniformly. The second edge is flipped (exchange s and t) with probability 1/2 and the edges are rewired to $u \sim t$ and $s \sim v$; the flip makes the two possible rewirings equally likely, since the stored order of the endpoints of an edge is arbitrary. Attempts that are not valid are skipped, but still count towards the QE attempts. Every replicate starts again from the original network.

Which switchings preserve the degree sequence? Both rewirings of two edges $u \sim v$, $s \sim t$, namely $\{u \sim t, s \sim v\}$ and $\{u \sim s, v \sim t\}$, preserve the degree sequence, because each vertex loses and gains exactly one edge.

Which of them produce forbidden edges? A rewiring creates a loop if the two new endpoints coincide, e.g. $u = t$ (or $s = v$) for $\{u \sim t, s \sim v\}$. It creates a multiedge if a new edge already exists in the graph. Edges that share a vertex (e.g. $u = s$) are the typical case: the rewiring then either reproduces an existing edge or does nothing useful, and is rejected by these two checks. Choosing the same edge twice is also rejected.

Choice of Q. By the coupon collector argument, about $\ln E$ rounds are needed for every edge to have been touched at least once, so we use $Q = \max(10, \lceil \ln E \rceil)$ (this gives $Q = 11$ for Hungarian, 12 for Basque, Chinese and Greek, 13 for Arabic, Catalan, English, Italian and Turkish, and 15 for Czech). [TODO: Report the fraction of attempts that were accepted.]

4.4 Exact computation with early stopping

To decide whether $\mathcal{C}_{NH} \geq \mathcal{C}$ it is not always necessary to compute $\mathcal{C}_{NH}$ fully. Vertices are visited in a uniformly random order, and after M vertices bounds on $\mathcal{C}_{NH}$ are available (we write S_M

for the sum of the computed $\mathcal{C}_i$):

$$\mathcal{C}_{NH}^{\min} = \frac{S_M}{N} \qquad \text{(unvisited vertices contribute 0)}, \tag{2}$$

$$\mathcal{C}_{NH}^{\max} = \frac{1}{N}\Big(S_M + \sum_{i \text{ unvisited}} u_i\Big), \qquad u_i = \frac{k_i + (N-1-k_i)/2}{N-1}. \tag{3}$$

The upper bound is tighter than the generic $\mathcal{C}_i \le 1$: a vertex of degree k_i has at most k_i vertices at distance 1 and every other vertex is at distance at least 2 (and unreachable vertices count as 0). Isolated vertices have $u_i = 0$ and are never visited. The search stops with "$\ge$" as soon as $\mathcal{C}_{NH}^{\min} \ge \mathcal{C}$ and with "$<$" as soon as $\mathcal{C}_{NH}^{\max} < \mathcal{C}$. This gives exactly the same answer as the full computation. It saves much work when $\mathcal{C}_{NH}$ is clearly below $\mathcal{C}$, but little when it is above (nearly all vertices must be visited before the lower bound reaches $\mathcal{C}$). The shortcut for degree-one vertices ($d_{ij} = d_{kj} + 1$) was not used: it requires the BFS of the neighbour k anyway.

4.5 Approximate computation

Exact computation costs about $N \cdot E$ operations per graph, which is not feasible for the largest networks (e.g. Czech: $N = 215082$, $E = 1.9 \cdot 10^6$). For those we estimate

$$\mathcal{C} \approx \frac{1}{M}\sum_{i=1}^{M} \mathcal{C}_i \tag{4}$$

over M vertices chosen uniformly at random for each random graph (no early stopping). The mean closeness of the *real* network is always computed exactly (one BFS from every vertex, parallelised over 8 threads), so the only sampling error comes from the random graphs. The cost is proportional to M/N.

Languages with $N \cdot E < 10^{10}$ (Arabic, Basque, Catalan, Chinese, Greek, Hungarian) were processed exactly (with early stopping). English, Italian and Turkish used $M/N = 10\%$ and Czech $M/N = 1\%$. To assess the effect of the approximation, Hungarian was run exactly and with decreasing M/N (Table **??**). The p-value for the binomial model is 0 in all cases, but the switching p-value decreases as M/N decreases, i.e. the approximation is biased towards smaller p-values.

M/N	1 (exact)	0.5	0.2	0.1	0.05	0.01
p-value (switching)	1.00	1.00	0.99	0.96	0.86	0.81

Table 3: Hungarian, $T = 100$: switching p-value as a function of the fraction of vertices used to estimate $\mathcal{C}_{NH}$.

4.6 Reproducibility

Each replicate uses its own pseudo-random stream seeded from the global seed and the replicate index, so results do not depend on the number of threads. The seed is 1 for all runs. T, Q and M/N per language are given above and in Table **??**. Everything is reproduced from the repository root with

```
Rscript code/build_networks.R ../Lab2/ud-treebanks-v2.18 data   # networks
make                                                            # compile
./run_all.sh                                                    # Table 2
./validate_approx.sh Hungarian 100                              # Table 3
```

[TODO: Fraction of accepted switching attempts, if measured.]

References

[1] M. Newman. *Networks: an introduction*. Oxford University Press, 2009.

[2] R. Ferrer-i-Cancho, R. V. Solé, R. Köhler. Patterns in syntactic dependency networks. *Physical Review E*, 69(5):051915, 2004.